

Threshold Reporting Protocol for Traceability in Anonymous Social Networks

Olivier Blazy¹[0000–0001–6205–8249] and Lola-Baie
Mallordy¹[0009–0005–1308–4412]

LIX, École Polytechnique, Institut Polytechnique de Paris
Inria, GRACE

Abstract. As anonymous social networks continue to grow in popularity, they raise serious challenges around abuse, misinformation, and harmful behavior. While traditional deanonymization techniques can address misuse, they often come at the cost of user privacy. Protecting honest users’ privacy while keeping malicious ones accountable remains a complex challenge. A promising alternative is *conditional deanonymization*, where anonymity is lifted only when abuse is collectively reported and verified by an authority.

In this work, we present a new cryptographic framework for conditional deanonymization that preserves user anonymity unless a threshold number of validated abuse reports is reached. Our design leverages threshold cryptography to ensure that deanonymization can only be triggered under well-defined, collectively enforced conditions—resisting false reports, misuse, and collusion. We formalize the model and provide a generic construction from standard cryptographic primitives, proving strong security guarantees even in the presence of adversarial authorities or malicious users. To illustrate its viability, we also propose a concrete instantiation based on lattice-based assumptions, making it a compelling candidate for post-quantum settings.

Keywords: Traceability · Signature · Privacy · Protocols · Anonymity.

1 Introduction

Social networks and online messaging services have become essential tools for everyday communication, playing an increasingly central role in our societies. To meet users’ expectations for privacy, many platforms strive to offer both *anonymity* and *confidentiality*. However, while anonymity can empower free expression, it may also foster a sense of impunity, shielding individuals who engage in illegal or harmful behavior. Regardless of one’s position on it, deanonymization for investigating malicious content is part of today’s digital landscape, regulated by laws and standards. Ignoring this reality can lead to significant risks, including unchecked criminal activity or disproportionate privacy intrusions. Solutions that guarantee anonymity without any possibility of accountability are often deemed unlawful or unsuitable, regardless of their technical merits. We adopt

an alternative approach: we analyze deanonymization requirements and provide technical means to fulfill them while still maximizing user privacy.

Aside from ethical concerns, many jurisdictions also require platforms to monitor illegal activities occurring within their services. For instance, the proposed EARN IT Act in the United States [17] aims to hold internet service providers accountable if they fail to remove child sexual abuse material, enabling legal action in cases of non-compliance. Similarly the Digital Services Act (DSA) [8] in Europe mandates that platforms implement clear mechanisms for users and authorities to flag illegal content. This leads to a central challenge:

how can a platform effectively moderate unlawful content without compromising the privacy of law-abiding users?

Granting platforms too much control over moderation risks abuse of power—where legitimate voices might be silenced, or doxxed for political, religious, or economic reasons. Striking the right balance between law enforcement and individual rights is both technically and ethically complex. We focus on minimizing the *false positives* (sanctioning a healthy post) and leave the *false negatives* (undetected unlawful behaviour) for future work. We aim for the following properties:

1. **User Privacy:** Deanonymization is strictly limited in scope to the post specified in the (multiple) reports. Authorities may not investigate a post deemed healthy by a majority of users.
2. **Deanonymization Security:** The process must remain confidential and undetectable by the targeted user or unauthorized third parties. Identifying information can then be provided promptly, securely, and *provably*.

To mitigate the risk of censorship, we propose *distributing moderation power* between the platform and its users. In our approach, anonymity is only lifted if a sufficient number of users report a post—essentially voting to revoke the poster’s anonymity—after which the platform may take action. This ensures that no single authority can deanonymize a user without collective input, reinforcing accountability without sacrificing privacy. Notably, our system is particularly effective in identifying *viral* content like fake news, which by nature reaches wide audiences and triggers several reports.

1.1 Related Work

Several privacy-preserving moderation protocols have been proposed in the last few years, in the case of end-to-end encrypted (E2EE) messaging systems [32,24], [19,13,3]. Such protocols aim to maintain confidentiality and anonymity for legitimate users while detecting (only) illegal content. Another sought-after property is the ability to trace the original sender of the illegal content (*traceability*) [32], which have often been shared and transferred multiple times; for instance in cases of hate speech or viral fake news.

E2EE setting. The first solution to be proposed [32] raises some privacy issues as it reveals the entire forwarding path (containing potentially honest users) of a reported message. Additionally, it introduces a large storage overhead

on the server side: for each message, the server must keep a 32 B token as long as the message can be traced. The subsequent work of [24] gets rid of this storage overhead, and forwarding path issue. However, their security relies on the *discrete logarithm problem*, hence their scheme is not robust against quantum attackers. Another solution [19] relies on *perceptual hash matching* [33] (PHM) to limit false negative reports. PHM allows, when a message is sent, to compute its hash value and compare it to known hashes of illegal content. If the hash of the message is close to one of these values, the message is very likely to be illegal. However, this method falls short from a privacy perspective, since it allows abusive servers to deanonymize any user. The work of [13] takes a different approach with a metadata-hiding message franking protocol. They use secret sharing in the message franking process such that the moderator can never see the reported messages, but only shares of it. However, their threat model is still limited by the assumption that the moderator is not malicious. The solution of [3] avoids all of these risks of abusive moderation with *pre-constrained encryption* [1]. This technique only decrypts messages that contain, for example, a word from a previously given list of keywords. However, this method can be easily avoided once the illegal users learn the list of forbidden keywords, which would require the list to be frequently updated.

Social network setting. Moderation solutions for social networks, where messages are shared publicly, differ from those in E2EE systems since users privacy lies in hiding the source of a message rather than its content. A survey was recently published regarding fake news detection techniques [31]. The authors go through the different classification of deep learning and machine learning models of text and image-based fake news such as [30,28,29]. However, these techniques will produce false positives and can cause mistrust due to their lack of explainability.

Threshold moderation protocols. To mitigate false positives, the first threshold reporting scheme was proposed [21] in the case of anonymous E2EE messaging, which requires a threshold of user complaints to trigger deanonymization. However, their solution is probabilistic and has a 10% theoretical error rate, hence is only suitable for relatively large thresholds. Moreover, their structure becomes “filled up” once too many complaints have been registered, and must be reset periodically. More recently in the same setting, another threshold reporting protocol was proposed by [4], this time non-probabilistic. Their scheme makes use of another server: their security properties rely on the assumption that the two servers will not collude. Finally, the work of [23] also tackles threshold moderation mechanism, but from another point of view: a user can break the platform policies up to t times, before having its identity revealed.

1.2 Contributions

We propose a threshold moderation protocol inspired by *group signatures*. Group signatures allow users to anonymously sign messages within a crowd of users. However, a trapdoor allows an authority (opener) to trace back a signature to

its original signer. Our solution distributes this trapdoor among users and the opener, using threshold decryption¹. We design a *Distributed Group Signature with Signature Dependent Opening* (DGSSDO) scheme. Let k be the opening threshold. Our scheme differs from group signature scheme in the following way:

- The opener can reveal a signer’s identity *only with the cooperation of k users*.
- The opening is *signature-dependent*: the opener requires the help of k users for each new signature, even on the same message.

We work in the trusted setup, where a trusted authority called the issuer is responsible for generating secret keys for any new member that joins the social network. Our scheme allows users to sign their messages anonymously, by producing a zero-knowledge proof (ZKP) of knowledge of their secret key. The signature also contains the user’s identity encrypted under two layers of encryption. To report a message, a user computes a signature-dependent token and sends it to the opener. Once the opener has collected enough tokens from different users, it can derive a signature-dependent decryption key. Using this key and its opening key, the opener can remove the two layers of encryption and recover the signer’s identity. However, any subset of less than k tokens does not leak any information. Additionally, we note that the tokens do not give any information on the users issuing the complaints.

We provide a generic construction using simple primitives (encryption, threshold decryption and signature schemes, as well as a zero-knowledge proof system). We prove its security, namely the following properties:

- *Traceability*: the opener with the help of k users can always recover the identity of the signer of a valid signature.
- *Token anonymity*: any collusion of the opener with less than k users cannot learn any information on the identity of the signer of a valid signature.
- *Opener anonymity*: any collusion of malicious users cannot learn anything on the signer’s identity of any message without the opener. We stress that this includes compromised identities: even if an adversary learns the secret keys of a user, it will not be able to identify any signature produced by it.

For all these properties, we consider strong adversaries that can corrupt adaptively users, while making signature or signature opening oracle queries. In our setting the threshold k is a constant, but can be updated very efficiently by updating only the keys used to generate the tokens (threshold decryption keys).

Since our protocol relies on users reporting abusive content while ignoring legitimate messages, it remains vulnerable to *collusion between the platform and a large group of malicious users*. For instance, if the platform were to generate

¹ For simplicity, we assume the website and legal enforcement authorities share one key. A secret sharing to ensure both parties are required for deanonymization could be considered, this would require extra steps at setup to ensure the social network does not view the unshared key. While easily doable with classical techniques, this would go beyond the scope of the paper and unnecessarily complexify our description.

enough fake accounts or bots, it could trigger the deanonymization of any target user. However, we believe bot detection to be a well-studied problem with existing solutions [9,15], which goes beyond the scope of this paper.

Lattice-based instantiation. We propose an instantiation of our protocol relying on both Module-LWE (MLWE) and Module-SIS (MSIS), providing post-quantum security. We produce signatures using the LaBRADOR SNARKs [6], that are sound under the MSIS assumption and can be made zero-knowledge under the MLWE assumption. We obtain small signature sizes: our signature size is comparable to LaBRADOR zk-SNARKs, which are around 60 KB.

Outline. The paper is organized as follows. In Section 2, we introduce the cryptographic primitives that underpin our construction. Section 3 formalizes our model for a DGSSDO scheme, along with its desired security properties. Then, we present a generic construction and detailed security proof in Section 4. Finally, in Section 5, we propose a concrete instantiation based on lattice assumptions.

2 Preliminaries

Notation. Sampling an element x uniformly from a set S is denoted $x \leftarrow \$ S$. By $\mathcal{A}$, we denote the adversary of a security experiment, which is a probabilistic polynomial-time algorithm. By $[n]$, we denote the set $\{1, \dots, n\}$.

2.1 Group Signatures

Group signatures allow users to sign messages while retaining anonymity within a crowd of users. However, misbehaving group members cannot remain unpunished as an authority, called the opening authority, is capable of tracing back a signature to its original signer. In a dynamic scheme, the issuer is a trusted authority responsible for providing any new group member with a set of keys.

Definition 1 (GS [5]). *A dynamic Group Signature scheme GS consists of the following algorithms:*

$\text{GS.KeyGen}(\lambda) \rightarrow \text{gpk}, \text{ok}, \text{isk}$: *On input the security parameter λ , returns the group public key gpk, the opening key ok and the issuer's key isk.*

$\text{GS.Join}(\text{isk}, i) \rightarrow \text{gsk}_i$: *using the issuer's key isk, returns a group signing key gsk_i for identity i .*

$\text{GS.Sign}(\text{gpk}, \text{gsk}_i, m) \rightarrow \sigma$: *returns a group signature σ on the message m .*

$\text{GS.Verif}(\text{gpk}, \sigma, m) \rightarrow \perp/\top$: *outputs $\top$ if σ is a valid signature for the message m , otherwise $\perp$.*

$\text{GS.Open}(\text{gpk}, \text{ok}, m, \sigma) \rightarrow \pi, i \in [n]/\perp$: *either returns the identity of the signer of σ and a proof π or returns $\perp$.*

To balance out the power given to the authority, group signature with message dependent opening (GSMDO) [26] schemes were introduced. The identity of a signer can only be determined from two pieces of information: the opener's key ok and a message-specific token delivered by a separate authority called the admitter. Neither authority is able to trace any signature alone.

2.2 Threshold protocols

Our generic construction relies on threshold decryption, to distribute the admit-ter's power amongst the users with respect to a given threshold t .

Definition 2 (TD [11]). A dynamic Threshold Decryption scheme TD consists of the following algorithms:

- TD.Setup(λ, t) $\rightarrow$ pk, isk: On input the security parameter λ , returns a public key pk and the issuer's key isk.
- TD.Join(isk, i) $\rightarrow$ sk _{i} : using the master secret key isk, returns a decryption key sk _{i} for user i .
- TD.Enc(m, pk) $\rightarrow$ c : returns an encryption c of the message m .
- TD.PartDec(c, sk_i) $\rightarrow$ p_i : outputs a partial decryption p_i of c .
- TD.FinDec($p_1, \dots, p_t$) $\rightarrow$ m : from t partial decryptions from t distinct users, outputs a final decryption such that for any $c \leftarrow \text{TD.Enc}(m, \text{pk})$, for all subset of t users $\mathcal{I}$,

$$\text{TD.FinDec}(\{\text{TD.PartDec}(c, \text{sk}_i)\}_{i \in \mathcal{I}}) = m.$$

To build such a scheme, we use a dynamic version of the generalization of Shamir's secret sharing to rings and modules [14].

Definition 3 (Generalized Shamir's scheme). The generalized Shamir's secret sharing scheme over a ring R is given as follows:

- SS.Setup($\lambda, t, s \in R$) $\rightarrow$ $f \in R[X]$: sample $a_1, \dots, a_{t-1} \leftarrow_{\$} R$ and return $f(X) = s + \sum_{j=1}^{t-1} a_j X^j$.
- SS.GenShare(f, x_i) $\rightarrow$ s_i : if $x_i - x_j$ is invertible in R for all previous share-holders $i \neq j$, return the share $s_i \leftarrow f(x_i)$. Otherwise, return $\perp$.
- SS.Rec($s_1, \dots, s_t$) $\rightarrow$ s : return

$$s \leftarrow \sum_{i=1}^t s_i \ell_i, \quad \text{for } \ell_i = \prod_{j \neq i} \frac{-x_j}{x_i - x_j}.$$

2.3 Lattice assumptions

The security of the primitives used in our protocols relies on the well-known computational lattice problems Module-LWE (MLWE) and Module-SIS (MSIS) [20,12]. For a degree d a power of 2 and an odd modulus q , denote R and R_q respectively to be the rings $R = \mathbb{Z}[x]/(X^d + 1)$ and $R_q = \mathbb{Z}_q[x]/(X^d + 1)$.

Definition 4 (MSIS). Given $A \in R_q^{k \times m}$, the MSIS problem with parameters $k, m > 0$ and $0 < B < q$ asks to find $z \in R_q^m$ such that $Az = 0$ and $0 < \|z\| \leq B$. An adversary $\mathcal{A}$ is said to have advantage ε in solving $\text{MSIS}_{k,m,B}$ if

$$\Pr [0 < \|z\| \leq B \wedge Az = 0 \mid A \leftarrow R_q^{k \times m}, z \leftarrow \mathcal{A}(A)] \geq \varepsilon.$$

Definition 5 (MLWE). The MLWE problem for $m, \lambda > 0$ and an error distribution χ over R asks the adversary $\mathcal{A}$ to distinguish between (1) $(A, As + e)$ for $A \leftarrow \$ R_q^{m \times \lambda}$, a secret $s \leftarrow \chi^\lambda$ and error $e \leftarrow \chi^m$, and (2) $(A, b) \leftarrow \$ R_q^{m \times \lambda} \times R_q^m$. Then $\mathcal{A}$ is said to have advantage ε in solving $\text{MLWE}_{m, \lambda, \chi}$ if

$$\begin{aligned} & \left| \Pr [b' = 1 \mid A \leftarrow \$ R_q^{m \times \lambda}, s \leftarrow \chi^\lambda, e \leftarrow \chi^m, \right. \\ & \quad \left. b' \leftarrow \mathcal{A}(A, As + e) \right] \\ & - \Pr [b' = 1 \mid A \leftarrow \$ R_q^{m \times \lambda}, b \leftarrow \$ R_q^m, b' \leftarrow \mathcal{A}(A, b)] \right| \geq \varepsilon. \end{aligned}$$

2.4 The Principal Relation

The LaBRADOR [6] proof system allows to prove the following principal relation over R_q , which can reduce to a rank-1 constraint system (R1CS) [6, Section 7]. Let $\text{ct}(f)$ be the constant coefficient of $f \in R_q$.

Definition 6 (Principal Relation). Let $\beta > 0$. Let $\mathcal{F}$ and $\mathcal{F}'$ be families of quadratic dot product functions $f: R_q^n \times \dots \times R_q^n \rightarrow R_q$ of the form

$$f(s_1, \dots, s_r) = \sum_{i,j=1}^r a_{ij} \langle s_i, s_j \rangle + \sum_{i=1}^r \langle r_i, s_i \rangle - b$$

for $a_{ij} \in R_q$ and $r_i \in R_q^n$. A witness for the principal relation $\mathcal{R}$ consists of $(s_1, \dots, s_r) \in (R_q^n)^r$ such that $\sum_{i=1}^r \|s_i\|_2^2 \leq \beta^2$ and

$$\forall f \in \mathcal{F}, f(s_1, \dots, s_r) = 0 \in R_q, \quad \forall f \in \mathcal{F}', \text{ct}(f(s_1, \dots, s_r)) = 0 \in \mathbb{Z}_q.$$

3 Our Model

Let $t \geq 2$ be the opening threshold. We describe our model as well as its different security properties.

3.1 Description

We define formally our model for a GSMDO scheme.

Definition 7. A Dynamic Distributed Group Signature with Signature Dependent Opening DGSSDO scheme consists of the following five algorithms:

DGSSDO.Setup(λ, t) $\rightarrow$ gpk, ok, isk: On input the security parameter λ and t the threshold for de-anonymisation, returns the group public key gpk, the opening key ok and the issuer's key isk.

DGSSDO.Join(isk, i) $\rightarrow$ (gsk $_i$, tsk $_i$): using the issuer's key isk, returns a group signing key gsk $_i$ and a token generation key tsk $_i$ for identity i .

DGSSDO.Sign(gpk, gsk $_i$, m) $\rightarrow$ σ : returns a group signature σ on m .

DGSSDO.Verif(gpk, σ , m) $\rightarrow$ $\perp/\top$: outputs $\top$ if σ is a valid signature for the message m , otherwise $\perp$.

$\text{DGSSDO.Token}(\text{gpk}, \text{tgk}_i, \sigma) \rightarrow t_\sigma^i$: returns a token t_σ^i for the signature σ .
 $\text{DGSSDO.Open}(\text{gpk}, \text{ok}, m, \sigma, \{t_\sigma^i\}_{i=1}^t) \rightarrow i \in [n]/\perp$: either returns the identity of the signer of σ or returns $\perp$.

The scheme must verify the following correctness property.

Definition 8 (Correctness). A DGSSDO scheme is correct if for any $t \in \text{poly}(\lambda)$, $\forall \text{gpk}, \text{ok}, \text{isk} \leftarrow \text{Setup}(\lambda, t)$, $\forall i \in [n], n \geq t$, $\forall (\text{gsk}_i, \text{tgk}_i) \leftarrow \text{Join}(\text{isk}, i)$, $\forall m \in \{0, 1\}^*$, we have

$$\begin{aligned}
 \text{Verif}(\text{gpk}, \text{Sign}(\text{gpk}, \text{gsk}_i, m), m) &= \top, \\
 \text{Open}(\text{gpk}, \text{ok}, m, \sigma, \{\text{Token}(\text{gpk}, \text{tgk}_i, m, \sigma)\}_{i=1}^t) &= \top.
 \end{aligned}$$

3.2 Security model

As in classical dynamic group signatures, a DGSSDO scheme must provide *traceability* and *anonymity*. As we split the opening capability between the opener and several users, we consider both *opener anonymity* and *token anonymity*. In the following, ε designates a negligible function in the security parameter λ .

Adaptive Traceability. In this game (see Figure 1), the adversary must forge a signature that is accepted by Verif , but that either cannot be opened or opens to the identity of a honest user. The adversary can choose adaptively which users to corrupt and query for honest signatures, but then the signature in its final answer must not open to these users.

Definition 9 (Traceability). A DGSSDO scheme has traceability if for any $t \in \mathbb{N}^*$ and adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{Trace}}(\mathcal{A}) := \Pr [\text{Exp}_{\mathcal{A}}^{\text{Trace}}(\lambda, t) = 1] \leq \varepsilon.$$

Token anonymity. This game is an indistinguishability game under adaptive chosen ciphertext attack. The adversary is given the opening key and can corrupt adaptively $t-1$ users. The adversary proposes a message and two identities. The challenger signs the given message with one of the two identities, and to win the adversary must be able to guess which identity was chosen by the challenger. The adversary has oracle access to tokens, for signatures different from the signature challenge. Mind that the adversary can choose corrupted identities for the signature challenge.

Definition 10 (Token anonymity). A DGSSDO scheme has token anonymity if for any $t \in \mathbb{N}^*$ and any adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{Token}}(\mathcal{A}) := |\Pr [\text{Exp}_{\mathcal{A}}^{\text{Token}}(\lambda, t) = 1] - 1/2| \leq \varepsilon.$$

$\text{Exp}_{\mathcal{A}}^{\text{Trace}}(\lambda, t):$	$\text{Exp}_{\mathcal{A}}^{\text{Open}}(\lambda, t):$
1 : $\text{gpk}, \text{ok}, \text{isk} \leftarrow \text{Setup}(\lambda, t)$ 2 : $\text{Corr} \leftarrow \emptyset, K \leftarrow \text{ok}$ 3 : $\text{continue} \leftarrow 1$ 4 : while (continue): 5 : $j, \text{continue} \leftarrow \mathcal{A}^{\text{Sign}}(\text{gpk}, K)$ 6 : $(\text{tgk}_j, \text{gsk}_j) \leftarrow \text{Join}(\text{isk}, j)$ 7 : $\text{Corr} \leftarrow \text{Corr} \cup \{j\}$ 8 : $K \leftarrow K \cup \{\text{tgk}_j, \text{gsk}_j\}$ 9 : $m, \sigma \leftarrow \mathcal{A}^{\text{Sign}}(K)$ 10 : $t_\sigma \leftarrow \{\text{Token}(\text{gpk}, \text{tgk}_i, \sigma)\}_{i \in [t]}$ 11 : return $(\text{Verif}(\text{gpk}, \sigma, m)) \wedge$ $((\text{Open}(\text{gpk}, \text{ok}, m, \sigma, t_\sigma) = \perp) \vee$ $(\text{Open}(\text{gpk}, \text{ok}, m, \sigma, t_\sigma) = (\pi, j) \wedge$ $(j \notin \text{Corr}) \wedge$ $(j, m) \text{ not queried to } \mathcal{O}_{\text{Sign}}))$	1 : $\text{gpk}, \text{ok}, \text{isk} \leftarrow \text{Setup}(\lambda, t)$ 2 : $\text{Corr} \leftarrow \emptyset, K \leftarrow \emptyset$ 3 : $\text{continue} \leftarrow 1$ 4 : while continue: 5 : $j, \text{continue} \leftarrow \mathcal{A}^{\text{Open}}(\text{gpk}, K)$ 6 : $(\text{tgk}_j, \text{gsk}_j) \leftarrow \text{Join}(\text{isk}, j)$ 7 : $\text{Corr} \leftarrow \text{Corr} \cup \{j\}$ 8 : $K \leftarrow K \cup \{\text{tgk}_j, \text{gsk}_j\}$ 9 : $m, j_0, j_1 \leftarrow \mathcal{A}(K)$ 10 : $b \leftarrow_{\$} \{0, 1\}$ 11 : $\sigma^* \leftarrow \text{Sign}(\text{gpk}, \text{gsk}_{j_b}, m)$ 12 : $b' \leftarrow \mathcal{A}^{\text{Open}}(\sigma^*)$ 13 : return $(b = b')$
Oracle $\mathcal{O}_{\text{Token}}(i, \sigma) :$	$\text{Exp}_{\mathcal{A}}^{\text{Token}}(\lambda, t):$
1 : if $\sigma = \sigma^*$ return $\perp$ 2 : if user i not created yet: 3 : $(\text{tgk}_i, \text{gsk}_i) \leftarrow \text{Join}(\text{isk}, i)$ 4 : return $\text{Token}(\text{gpk}, \text{tgk}_i, \sigma)$	1 : $\text{gpk}, \text{ok}, \text{isk} \leftarrow \text{Setup}(\lambda, t)$ 2 : $\text{Corr} \leftarrow \emptyset, K \leftarrow \text{ok}$ 3 : $\text{continue} \leftarrow 1$ 4 : while $\text{continue} \wedge \text{Corr} < t - 1 :$ 5 : $j \leftarrow \mathcal{A}^{\text{Token}}(\text{gpk}, K)$ 6 : $(\text{tgk}_j, \text{gsk}_j) \leftarrow \text{Join}(\text{isk}, j)$ 7 : $\text{Corr} \leftarrow \text{Corr} \cup \{j\}$ 8 : $K \leftarrow K \cup \{\text{tgk}_j, \text{gsk}_j\}$ 9 : $m, j_0, j_1 \leftarrow \mathcal{A}(K)$ 10 : $b \leftarrow_{\$} \{0, 1\}$ 11 : $\sigma^* \leftarrow \text{Sign}(\text{gpk}, \text{gsk}_{j_b}, m)$ 12 : $b' \leftarrow \mathcal{A}^{\text{Token}}(\sigma^*)$ 13 : return $(b = b')$
Oracle $\mathcal{O}_{\text{Open}}(\{t_\sigma^i\}_{i=1}^t) :$	
1 : if $\sigma = \sigma^*$ return $\perp$ 2 : return $\text{Open}(\text{gpk}, \text{ok}, m, \sigma, \{t_\sigma^i\}_{i=1}^t)$	
Oracle $\mathcal{O}_{\text{Sign}}(i, m) :$	
1 : if i not created yet: 2 : $(\text{tgk}_i, \text{gsk}_i) \leftarrow \text{Join}(\text{isk}, i)$ 3 : return $\text{Sign}(\text{gpk}, \text{gsk}_i, m)$	

Fig. 1: The security experiments $\text{Exp}_{\mathcal{A}}^{\text{Token}}$, $\text{Exp}_{\mathcal{A}}^{\text{Open}}$ and $\text{Exp}_{\mathcal{A}}^{\text{Trace}}$, and their oracles $\mathcal{O}_{\text{Token}}$, $\mathcal{O}_{\text{Open}}$ and $\mathcal{O}_{\text{Sign}}$.

Opener anonymity. This game is very similar to the previous game. This time, the adversary can corrupt adaptively any number of users, but does not know the opener's key. The adversary proposes a message and two identities. The challenger signs the given message with one of the two identities, and to win the adversary must be able to guess which identity was chosen by the challenger.

The adversary has access to an opening oracle $\mathcal{O}_{\text{Open}}$, for signatures different from the signature challenge.

Definition 11 (Opener anonymity). A DGSSDO scheme has opener anonymity if for any $t \in \mathbb{N}^*$ and any adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{Open}}(\mathcal{A}) := |\Pr[\text{Exp}_{\mathcal{A}}^{\text{Open}}(\lambda, t) = 1] - 1/2| \leq \varepsilon.$$

4 Generic Construction

We construct a DGSSDO scheme, and prove its security relying on the security of its underlying primitives.

4.1 Our protocol

Let Σ be a signature scheme with existential unforgeability under adaptive chosen-message attack (EU-ACMA) [16], E an INDCCA2 encryption scheme, TD a dynamic threshold decryption scheme with INDCCA2-TD security (indistinguishability under chosen plaintext attacks for threshold decryption [27]), and let $\mathcal{I}$ be the set of valid identities. Our generic construction is given in Figure 2.

Setup. In our scheme, the issuer has two sets of keys: a signature key sk_{Σ} and an issuer key isk_{msk} for a threshold decryption scheme. When a new member i wants to join the group, the issuer sends him a signature from sk_{Σ} on its identity i which will be its group signing key gsk_i , and a new threshold decryption key from isk_{msk} which will be its token generation key tgk_i . The opener's keys consists of a simple key pair $(\text{pk}_{\text{ok}}, \text{sk}_{\text{ok}})$ for a classic encryption scheme.

Let $\Pi = \{\text{Prove}, \text{Verif}\}$ be a non-interactive ZKP system for the following relation

$$\mathcal{R} = \left\{ \begin{array}{l} (\text{pk}_{\text{ok}}, \text{vk}, \text{pk}_{\text{msk}}), \\ (c', i, \text{gsk}_i, r, r') \end{array} \left| \begin{array}{l} c = \text{TD.Enc}(\text{pk}_{\text{msk}}, \text{bin}(\text{Enc}(\text{pk}_{\text{ok}}, i; r)); r') \\ i \in \mathcal{I} \\ \Sigma.\text{Verif}(\text{vk}, i, \text{gsk}_i) = \top \end{array} \right. \right\}. \quad (1)$$

Signing a message. When a user wants to sign a message m , its signature consists of an encryption of its identity under both the issuer's threshold public key pk_{msk} and the opener public key pk_{ok} . Then, the signer proves that the encryption was honest and its identity with Π . Finally, to ensure that the signature depends on the message, we add m to the transcript of the ZKP for the Fiat-Shamir transform (thus, we add m as input to the $\Pi.\text{Prove}$ and $\Pi.\text{Verif}$ algorithms). Then, verifying a signature only consists of verifying the zero-knowledge proof π with the public key gpk and the message m .

Opening a signature. To report a message, a user partially decrypts the ciphertext part c' with its threshold decryption key tgk_i . When the opener has gathered enough partial decryptions, it can reconstruct the ciphertext c that was encrypted under its public key pk_{ok} , and thus recover the signer's identity.

DGSSDO.Setup(λ, t):	DGSSDO.Sign($\text{gpk}, \text{gsk}_i, m$):
1 : $(\text{pk}_{\text{ok}}, \text{sk}_{\text{ok}}) \leftarrow \text{E.KeyGen}(\lambda)$	1 : $c \leftarrow \text{E.Enc}(i, \text{pk}_{\text{ok}})$
2 : $(\text{vk}, \text{sk}_{\Sigma}) \leftarrow \Sigma.\text{KeyGen}(\lambda)$	2 : $c' \leftarrow \text{TD.Enc}(c, \text{pk}_{\text{msk}})$
3 : $(\text{pk}_{\text{msk}}, \text{isk}_{\text{msk}}) \leftarrow \text{TD.Setup}(\lambda, t)$	3 : $\pi \leftarrow \Pi.\text{Prove}(m, c', i, \text{gsk}_i, r, r')$
4 : $\text{gpk} \leftarrow (\text{pk}_{\text{ok}}, \text{vk}, \text{pk}_{\text{msk}})$	4 : return $\sigma = (c', \pi)$
5 : $\text{isk} \leftarrow (\text{sk}_{\Sigma}, \text{isk}_{\text{msk}})$	DGSSDO.Verif($\text{gpk}, \sigma = (c', \pi), m$):
6 : return $\text{gpk}, \text{ok} = \text{sk}_{\text{ok}}, \text{isk}$	1 : return $\Pi.\text{Verif}(m, \text{gpk}, \pi)$
DGSSDO.Join($\text{isk}, i \in \mathcal{I}$):	DGSSDO.Open($\text{gpk}, \text{ok}, m, \sigma, \{t_{\sigma}^i\}_{i=1}^t$):
1 : $\text{gsk}_i \leftarrow \Sigma.\text{Sign}(\text{sk}_{\Sigma}, i)$	1 : $c \leftarrow \text{TD.FinDec}(\{t_{\sigma}^i\}_{i=1}^t)$
2 : $\text{tgk}_i \leftarrow \text{TD.Join}(\text{isk}_{\text{msk}}, i)$	2 : $i \leftarrow \text{E.Dec}(c, \text{ok})$
3 : return $(\text{tgk}_i, \text{gsk}_i)$	3 : if $i \in \mathcal{I}$: return i
DGSSDO.Token($\text{gpk}, \text{tgk}_i, \sigma$):	4 : else return $\perp$
1 : return $t_{\sigma}^i \leftarrow \text{TD.PartDec}(c', \text{tgk}_i)$	

Fig. 2: Generic construction of our DGSSDO scheme. Let $\mathcal{I}$ be the set of identities.

4.2 Security analysis

We prove the security of our generic construction under the security of the underlying signature scheme Σ , the threshold decryption scheme TD, the encryption scheme E and the zero-knowledge proof system Π .

Theorem 1. *Under the assumption that TD is INDCCA2-TD and Π is zero-knowledge, our scheme has token anonymity. More specifically, for any adversary $\mathcal{A}$ we have*

$$\text{Adv}_{\text{Token}}(\mathcal{A}) \leq \text{Adv}_{\text{ZK}}(\mathcal{A}) + \text{Adv}_{\text{INDCCA2-TD}}(\mathcal{A}).$$

Proof. We prove this theorem by a sequence of games.

Game G_0 : the original game $\text{Exp}_{\mathcal{A}}^{\text{Token}}(\lambda, t)$ instantiated with our scheme:

- The challenger runs the Setup to compute $\text{gpk} = (\text{pk}_{\text{ok}}, \text{vk}, \text{pk}_{\text{msk}}), \text{ok}, \text{isk} = (\text{sk}_{\Sigma}, \text{isk}_{\text{msk}})$ where $(\text{pk}_{\text{ok}}, \text{ok}) \leftarrow \text{E.KeyGen}(\lambda)$, $(\text{vk}, \text{sk}_{\Sigma}) \leftarrow \Sigma.\text{KeyGen}(\lambda)$ and $(\text{pk}_{\text{msk}}, \text{isk}_{\text{msk}}) \leftarrow \text{TD.Setup}(\lambda, t)$. It sends gpk, ok to the adversary.
- The adversary chooses adaptively parties to corrupt while $|\text{Corr}| < t$ to get their keys $\text{tgk}_i \leftarrow \text{TD.Join}(\text{isk}_{\text{msk}}, i)$, $\text{gsk}_i \leftarrow \Sigma.\text{Sign}(\text{sk}_{\Sigma}, i)$, and can make queries to $\mathcal{O}_{\text{Token}}$.
- The adversary chooses a message m and two identities $j_0, j_1 \in \mathcal{I}$.
- The challenger draws $b \leftarrow \{0, 1\}$ and computes a signature $\sigma^* = (c', \pi)$ on m with identity j_b . It sends the challenge σ^* to the adversary.
- The adversary can make some more queries to $\mathcal{O}_{\text{Token}}$ for any $\sigma \neq \sigma^*$ and outputs b' .

Game G_1 : In this game, the challenger computes π using the simulator of Π instead of the witnesses in the challenge phase. Hence, we have exactly

$$|\text{Adv}_0(\mathcal{A}) - \text{Adv}_1(\mathcal{A})| \leq \text{Adv}_{\text{ZK}}(\mathcal{A}).$$

Game G_2 : In this game, the challenger samples c' uniformly from the ciphertext space instead of setting $c' \leftarrow \text{TD.Enc}(c, \text{pk}_{\text{msk}})$ when computing the challenge signature. The adversary knows $|\text{Corr}| < t$ partial decryption keys and can send adaptively decryption queries to the oracle $\mathcal{O}_{\text{Token}}$, hence we have exactly

$$|\text{Adv}_1(\mathcal{A}) - \text{Adv}_2(\mathcal{A})| \leq \text{Adv}_{\text{INDCCA2-TD}}(\mathcal{A}).$$

In G_2 , all the adversary's inputs are independent of the challenge b , hence $\mathcal{A}$ has no better strategy than answering at random and $\text{Adv}_2(\mathcal{A}) = 0$. The theorem follows by chaining the inequalities.

Theorem 2. *Under the assumption that Π is zero-knowledge and E is INDCCA2, our scheme has opener anonymity. More specifically, for any adversary $\mathcal{A}$ we have*

$$\text{Adv}_{\text{Open}}(\mathcal{A}) \leq \text{Adv}_{\text{ZK}}(\mathcal{A}) + \text{Adv}_{\text{INDCCA2}}(\mathcal{A}).$$

Proof. We prove this theorem by a sequence of games.

Game G_0 : the original game $\text{Exp}_{\mathcal{A}}^{\text{Open}}(\lambda, t)$ instantiated with our scheme:

- The challenger runs the **Setup** to compute $\text{gpk} = (\text{pk}_{\text{ok}}, \text{vk}, \text{pk}_{\text{msk}}), \text{ok}, \text{isk} = (\text{sk}_{\Sigma}, \text{isk}_{\text{msk}})$ where $(\text{pk}_{\text{ok}}, \text{ok}) \leftarrow E.\text{KeyGen}(\lambda)$, $(\text{vk}, \text{sk}_{\Sigma}) \leftarrow \Sigma.\text{KeyGen}(\lambda)$ and $(\text{pk}_{\text{msk}}, \text{isk}_{\text{msk}}) \leftarrow \text{TD.Setup}(\lambda, t)$. It sends gpk to the adversary.
- The adversary chooses adaptively any number of parties to corrupt to get their keys $\text{tgk}_i \leftarrow \text{TD.Join}(\text{isk}_{\text{msk}}, i)$, $\text{gsk}_i \leftarrow \Sigma.\text{Sign}(\text{sk}_{\Sigma}, i)$, and can make queries to $\mathcal{O}_{\text{Open}}$.
- The adversary chooses a message m and two identities $j_0, j_1 \in \mathcal{I}$.
- The challenger draws $b \leftarrow_{\$} \{0, 1\}$ and signs m with identity j_b . It sends the challenge signature $\sigma^* = (c', \pi)$ to the adversary.
- The adversary makes some more queries to $\mathcal{O}_{\text{Open}}$ for any signature $\sigma \neq \sigma^*$ and outputs b' .

Game G_1 : In this game, the challenger computes π using the simulator of the proof system instead of the witnesses in the challenge phase. Hence, we have exactly

$$|\text{Adv}_0(\mathcal{A}) - \text{Adv}_1(\mathcal{A})| \leq \text{Adv}_{\text{ZK}}(\mathcal{A}).$$

Game G_2 : In this game, the challenger samples c uniformly from the ciphertext space instead of setting $c \leftarrow E.\text{Enc}(j_b, \text{pk}_{\text{ok}})$ when computing the challenge signature. The adversary can send adaptively decryption queries to the oracle $\mathcal{O}_{\text{Open}}$, hence we have exactly

$$|\text{Adv}_1(\mathcal{A}) - \text{Adv}_2(\mathcal{A})| \leq \text{Adv}_{\text{INDCCA2}}(\mathcal{A}).$$

In G_2 , all the adversary's inputs are independent of the challenge bit $b \in \{0, 1\}$, hence $\text{Adv}_2(\mathcal{A}) = 0$. The theorem follows by chaining the reductions.

Theorem 3. Let q_{Sign} be the number of queries to the signing oracle. Under the assumption that Π is sound and zero-knowledge, and Σ is EU-ACMA, our scheme has traceability. For any adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{Trace}}(\mathcal{A}) \leq \frac{q_{\text{Sign}} \cdot \text{Adv}_{\text{ZK}}(\mathcal{A}) + \text{Adv}_{\text{EU-ACMA}}(\mathcal{A})}{1 - \text{Adv}_{\text{SND}}(\mathcal{A})}.$$

Proof. We prove this theorem by a reduction to the EU-ACMA security of the signature scheme Σ . Let $\mathcal{A}$ be an adversary with probability ε to break the traceability property of our scheme.

The traceability game:

- The challenger runs the **Setup** to compute $\text{gpk} = (\text{pk}_{\text{ok}}, \text{vk}, \text{pk}_{\text{msk}}), \text{ok}$ and $\text{isk} = (\text{sk}_{\Sigma}, \text{isk}_{\text{msk}})$ where $(\text{pk}_{\text{ok}}, \text{ok}) \leftarrow \text{E.KeyGen}(\lambda)$, $(\text{vk}, \text{sk}_{\Sigma}) \leftarrow \Sigma.\text{KeyGen}(\lambda)$ and $(\text{pk}_{\text{msk}}, \text{isk}_{\text{msk}}) \leftarrow \text{TD.Setup}(\lambda, t)$. It sends gpk, ok to the adversary.
- The adversary chooses adaptively parties to corrupt, and queries to $\mathcal{O}_{\text{Sign}}$. When a party i is corrupted, the challenger runs the **Join** algorithm and sends the corrupted keys $\text{gsk}_i = \Sigma.\text{Sign}(\text{sk}_{\Sigma}, i)$ and $\text{tgk}_i = \text{TD.Join}(\text{isk}_{\text{msk}}, i)$ to the adversary.
- The adversary outputs a message m and a signature σ .

Reduction setup. The reduction $\mathcal{R}$ takes as input the verification key vk for the signature scheme Σ . It generates the other keys from the **Setup** as in the protocol. For any corruption request on an identity i , $\mathcal{R}$ asks its own signing oracle for a signature gsk_i on i , generates tgk_i faithfully and sends $(\text{gsk}_i, \text{tgk}_i)$ to $\mathcal{A}$. To answer $\mathcal{O}_{\text{Sign}}$ queries on a message m with identity $i \in \text{Hon}$, $\mathcal{R}$ computes faithfully c' , and uses the simulator of the proof system to compute π without the witness gsk_i . Let q_{Sign} the number of such queries, the probability for the adversary to distinguish such signatures from honest signatures is bounded by $q_{\text{Sign}} \cdot \text{Adv}_{\text{ZK}}(\mathcal{A})$.

Main reduction. Assume that $\mathcal{A}$ outputs a forgery $(m^*, \sigma^* = (c', \pi))$ which satisfies the requirements of the traceability game *i.e.*, π is a valid proof and c' does not decrypt as $i \in \text{Corr}$. The reduction exploits the forgery as follows. We use the extractor from the ZKP, which extracts a witness

$$w = (i^*, \text{gsk}_{i^*}, c, r, r')$$

from $\mathcal{A}$, such that gsk_{i^*} is a valid signature for vk on i^* with probability $1 - \text{Adv}_{\text{SND}}(\mathcal{A})$. Since the reduction only asked for signatures on corrupted identities to its signing oracle, it never asked for a signature on $i^* \notin \text{Corr}$. Thus, the probability $p_{\mathcal{R}}$ for the reduction to output a valid forgery (i^*, gsk_{i^*}) is at least

$$p_{\mathcal{R}} \geq \varepsilon(1 - \text{Adv}_{\text{SND}}(\mathcal{A})) - q_{\text{Sign}} \cdot \text{Adv}_{\text{ZK}}(\mathcal{A}),$$

and the theorem follows.

5 Our Lattice-based Instantiation

We instantiate our construction with lattice-based primitives. For readability, we focus on the token anonymity, and only propose an IND CPA version of the opener anonymity (we only use an IND CPA encryption for the opener).

Signature. We use the simple signature scheme Σ_{SUF} that can be derived from [10], which we recall in Figure 3. This scheme only has selective unforgeability, but can be turned into an EU-ACMA scheme Σ using chameleon hashes [18] (we recall the definition in Appendix A.1).

$\Sigma_{\text{SUF}}.\text{KeyGen}(\lambda)$:	$\Sigma_{\text{SUF}}.\text{Sign}(\text{sk}, m \in R_q)$:
1 : $A \leftarrow \$ R_p^{n \times (n+m)}, B' \leftarrow \$ R_p^{n \times \ell n}$	1 : $\sigma \leftarrow \text{Sample}([A \mid B + mG \mid B'], \text{sk}, u, s)$
2 : $R \leftarrow \$ S_1^{(n+m) \times \ell n}, u \leftarrow \$ R_p^n$	2 : return σ
3 : $B \leftarrow AR$	$\Sigma_{\text{SUF}}.\text{Verif}(\text{vk}, \sigma, m)$:
4 : $\text{vk} \leftarrow (A, B, B', u), \text{sk} \leftarrow R$	1 : return (σ is a short vector)
5 : return vk, sk	2 : $\wedge [A \mid B + mG \mid B']\sigma = u \pmod p$
$\text{TD}.\text{Setup}(\lambda, t)$:	$\text{TD}.\text{Enc}(\text{pk} = (A, b), m \in \{0, 1\}^d)$:
1 : $A \leftarrow \$ R_q^{N \times K}$	1 : $r \leftarrow \$ S_1^N$
2 : $s \leftarrow \$ S_1^K, e \leftarrow \$ S_1^N$	2 : $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \leftarrow \begin{bmatrix} A^t \\ b^t \end{bmatrix} r + \begin{bmatrix} 0 \\ \lfloor \frac{q}{2} \rfloor m \end{bmatrix} \pmod q$
3 : $b \leftarrow As + e \in R_q^N$	3 : return (c_1, c_2)
4 : $\text{pk} \leftarrow (A, b)$	$\text{TD}.\text{PartDec}(\text{sk}_i, c)$:
5 : $\text{isk} \leftarrow \text{SS}.\text{Setup}(\lambda, t, s)$	1 : return $p_i = c_2 - \text{sk}_i^t c_1 \pmod q$
6 : return pk, isk	$\text{TD}.\text{FinDec}(p_1, \dots, p_t)$:
$\text{TD}.\text{Join}(\text{isk}, i)$:	1 : $m' \leftarrow \text{SS}.\text{Rec}(p_1, \dots, p_t)$
1 : $\text{sk}_i \leftarrow \text{SS}.\text{GenShare}(\text{isk}, i)$	2 : return m with i -th bit 0 if
2 : return sk_i	the i -th bit of m' is closer to 0
	than $\lfloor q/2 \rfloor$, otherwise 1

Fig. 3: Top: the signature scheme Σ_{SUF} algorithms. Let p be a prime number, and $\text{Sample}(A, R, u, s)$ be the discrete gaussian sampler [22] that on input a trapdoor R for A , outputs v s.t. $Av = u \pmod p$ with standard deviation s . Bottom: the CPA threshold decryption scheme TD. Let q be a prime modulus and d the dimension of R_q over $\mathbb{Z}$.

We use the ring version of the SIS-based chameleon hash function from [7]:

- $\Sigma.\text{KeyGen}(\lambda)$: output $(\text{vk}, \text{sk}) \leftarrow \Sigma_{\text{SUF}}.\text{KeyGen}$ and a chameleon hash key $\text{ck} = (a_0, a_1) \in R_q^n \times R_q^m$. The public key is $\text{vk}' = (\text{vk}, \text{ck})$.
 $\Sigma.\text{Sign}(\text{sk}, m \in R_q^n)$: compute a hash $h = \text{CHash}(\text{ck}, m; r) = a_0 m + a_1 r \in R_q$ for a random $r \leftarrow \$ R_q^m$ and sign it as $\sigma \leftarrow \Sigma_{\text{SUF}}.\text{Sign}(\text{sk}, h)$. Output (σ, r) .

$\Sigma.\text{Verif}(\text{vk}, (\sigma, r), m)$: set $h \leftarrow \text{CHash}(\text{ck}, m; r)$ and output $\Sigma_{\text{SUF}}.\text{Verif}(\text{vk}, \sigma, h)$.

Threshold Regev encryption. We use a Regev encryption [25] for both the classic encryption and the threshold decryption. Indeed, Regev encryption can be turned into a threshold decryption scheme in a straightforward way, by secret sharing the decryption key between users with the generalized Shamir's secret sharing over rings and modules (see Definition 3). We get the following threshold scheme TD described in Figure 3.

Standard Regev encryption only provides IND CPA security. However, since our signature contains a ZKP that proves, among other statements, the integrity of the ciphertext c' , this automatically gives us an INDCCA2-TD threshold decryption scheme.

5.1 Our DGSSDO scheme

Lastly, it remains to detail how to compute the ZKP and how to get the CCA2 token anonymity. Let p be a prime number. The set of identities $\mathcal{I}$ is

$$\mathcal{I} = \{i \in R_p \mid i \text{ is binary} \wedge \|i\|_1 \leq \omega\}$$

where ω is a function of λ . Let $\text{bin}(\cdot)$ be the binary decomposition of a vector.

- $\text{DGSSDO.Sign}(\text{gpk}, \text{gsk}_i, m)$:
 $c = (c_1, c_2) \leftarrow \text{E.Enc}(i, \text{pk}_{\text{ok}}; r)$,
 $c' \leftarrow \text{TD.Enc}(\text{bin}(c_2), \text{pk}_{\text{msk}}; r')$ (we only encrypt c_2 for efficiency since c_1 is independent of the secret),
 $\pi \leftarrow \Pi.\text{Prove}(m, c', i, \text{gsk}_i, r, r')$ using LaBRADOR [6] for the relation $\mathcal{R}$ and adding m to the transcript for the Fiat-Shamir transform,
Return $\sigma \leftarrow (c_1, c', \pi)$.
- $\text{DGSSDO.Token}(\text{gpk}, \text{tgk}_i, \sigma = (c_1, c', \pi), m)$:
If $\Pi.\text{Verif}(m, \text{gpk}, \pi) = \perp$ return $\perp$, else return $\text{TD.PartDec}(c', \text{tgk}_i)$.

As we mentioned, the additional verification of the proof π in **Token** gives us INDCCA2 security since in particular, the relation $\mathcal{R}$ (1) includes the knowledge of a witness (m, r) such that $c' = \text{TD.Enc}(\text{pk}_{\text{msk}}, m; r)$.

5.2 Reduction to the principal relation.

We show that the relation $\mathcal{R}$ (1) proven by the signer with Π can be reduced to a principal relation (Definition 6), hence is compatible with LaBRADOR [6].

Let q_E be the prime modulus of the encryption scheme, p be the ℓ bits prime modulus for the signature scheme and $q = p \cdot p'$ be the composite proof modulus. With our instantiation, knowing a witness $w \in \mathcal{R}$ amounts to knowing

$$w = (\text{bin}(c_2), i, \text{gsk}_i, r, r')$$

such that:

1. $(c_1, c_2) = \text{Enc}(\text{pk}_{\text{ok}}, i; r)$ and $c' = \text{TD.Enc}(\text{pk}_{\text{msk}}, \text{bin}(c_2); r')$ *i.e.*,
 - (a) $\|r\| \leq B := \sqrt{Nd}$ and $\|r'\| \leq B$.
 - (b) $\text{bin}(c_2)$ is binary.
 - (c) Let $G_{\text{bin}} = I_n \otimes [1 \ 2 \ \dots \ 2^{\ell-1}]$ be a gadget matrix. Let $\text{pk}_{\text{ok}} = (A, b)$ and $\text{pk}_{\text{msk}} = (A', b')$ the opener and the issuer public keys. The vector

$$v = \frac{1}{q_E} \left(\begin{bmatrix} c_1 \\ 0 \\ c'_1 \\ c'_2 \end{bmatrix} - \begin{bmatrix} A & 0 & 0 & 0 \\ b^t & \lfloor \frac{q_E}{2} \rfloor & 0 & -G_{\text{bin}} \\ 0 & 0 & A' & 0 \\ 0 & 0 & b'^t & \lfloor \frac{q_E}{2} \rfloor \end{bmatrix} \begin{bmatrix} r \\ i \\ r' \\ \text{bin}(c_2) \end{bmatrix} \right)$$

verifies

$$\|v\| \leq (B\sqrt{Kd}/2 + 1)\sqrt{d(N + \log_2(q))}.$$

2. $i \in \mathcal{I}$ *i.e.*,
 - (a) $i \in \mathcal{R}_q$ is binary.
 - (b) $\|i\|_1 = \omega$.
3. $\Sigma.\text{Verif}(\text{vk}, i, \text{gsk}_i) = \top$ *i.e.*,
 - (a) The vector $(i, s_{i_1}, s_{i_2}, s_{i_3}, r_i)$ satisfies the following quadratic relation

$$p' [A \mid B + hG \mid B'] \begin{bmatrix} s_{i_1} \\ s_{i_2} \\ s_{i_3} \end{bmatrix} = p' [A \mid B \mid G \mid B'] \begin{bmatrix} s_{i_1} \\ s_{i_2} \\ h s_{i_2} \\ s_{i_3} \end{bmatrix} = p' u \pmod{q}.$$

where $h = a_0 i + a_1 r_i$ for $\text{ck} = (a_0, a_1)$ the chameleon hashing key.

- (b) For N_{Sign} the length of gsk_i and s the standard deviation of the discrete gaussian sampler, $\|\text{gsk}_i\| \leq s\sqrt{2N_{\text{Sign}}d}$ (see [2, Lemma 1.5]).

All of these statements are already encompassed by the principal relation except the statements (2a,1b) on binary messages and (2b) bounding the ℓ_1 -norm. We can prove that a vector x is binary with the principal relation by proving the quadratic relation $\langle x, x - 1 \rangle = 0 \pmod{q}$ and $\|x\| \leq B_{\text{bin}}$ such that the relation holds over the integers. As for the ℓ_1 -norm, let $\sigma^{-1} : R_q \rightarrow R_q$ be the automorphism $X \mapsto X^{-1}$. We can prove that $\|v\|_1 = \omega$ by proving the following evaluation at zero

$$\left(\sigma^{-1} \left(\sum_{j=0}^{d-1} X^j \right) \cdot v - \omega \right) (0) = 0 \pmod{q}.$$

5.3 Overhead estimates

The size of the LaBRADOR proof π is around 60 KB. For an encryption modulus $q_E = 3329$, degree $d = 128$ and $N = 4, K = 9$, the ciphertexts (c_1, c') add an overhead of 5.76 KB. This gives us a signature size of approximately 66 KB, comparable to the size of the zk-SNARK. We summarize the resulting signature, tokens and public key sizes in Table 1.

Note that the threshold can be updated by just renewing the TD keys (which merely consists of a secret sharing), which is not a costly operation.

Signature σ	gpk	Token t_σ^i	Secret key $(\text{tgk}_i, \text{gsk}_i)$
66 KB	79.4 KB	1.9 KB	7.8 KB

Table 1: Summary of the sizes of the different elements in our protocol, taking a signature modulus $p = 2^{38} - 107$, a matrix A in vk of size 2×3 , $\ell = 5$.

5.4 Security analysis

We show that we can apply the theorems from Section 4.2, except that we prove a weaker version of the opener anonymity (CPA-opener anonymity) where we remove access to the oracle $\mathcal{O}_{\text{Open}}$. To get the full opener anonymity, one would have to replace the classical Regev encryption on the opener’s side by an INDCCA2 encryption scheme.

We prove the security of the signature and threshold decryption schemes in Appendix A.2. The Regev encryption is IND CPA under the MLWE assumption [25], from which we derive the INDCCA2-TD security of the threshold decryption scheme under the MLWE and MSIS assumptions (Appendix A.2). Moreover, the LaBRADOR proof system can be made zero-knowledge under the MLWE assumption as shown in [6, Section 6], and is sound under the MSIS assumption [6]. We show the selective unforgeability of the signature Σ_{SUF} in Appendix A.2, which suffices to show the EU-ACMA security of Σ [18] (the chameleon hash function from [7] being secure under the MSIS assumption). Finally, we can apply the theorem from Section 4.2 and obtain the following result.

Theorem 4. *Our instantiation has token anonymity, CPA-opener anonymity, traceability and non-frameability under the MSIS and MLWE assumptions.*

6 Conclusion

Balancing user privacy with effective moderation remains a critical challenge for online platforms. Achieving a system that mitigates unlawful content while preserving user confidentiality is essential to building trust among users and complying with international legal frameworks. To address this, we present a novel *Distributed Group Signature with Signature Dependent Opening* scheme. This approach leverages a threshold of user reports to de-anonymize malicious actors selectively. The scheme’s security is rigorously analyzed and formally proven, assuming the security of the underlying cryptographic primitives.

Our work includes a lattice-based instantiation based on the MSIS and MLWE assumptions, utilizing state-of-the-art lattice protocols. We obtain small signature sizes, that are dominated by the size of the zk-SNARK from LaBRADOR [6], hence providing a first reasonable post-quantum solution.

A Appendix

A.1 Additional Background

We use chameleon hashes [18] to turn a selective unforgeable signature scheme into an EU-ACMA scheme. A chameleon hash is similar to a hash function, but with a secret trapdoor that allows arbitrary to find collision.

Definition 12 (Chameleon Hash Function [18]). *A Chameleon Hash Function is defined by the following algorithms:*

- $\text{CH.KeyGen}(\lambda) \rightarrow (\text{ck}, \text{tk})$: outputs the chameleon hash key ck and the trapdoor tk .
- $\text{CHash}(\text{ck}, m; r) \rightarrow h, d$: picks a random r if it is not given, and outputs a chameleon hash h .
- $\text{Coll}(\text{ck}, m, r, m', \text{tk}) \rightarrow r$: takes as input the public key ck , the trapdoor tk , a start message m and randomness r , a target message m' and outputs a randomness r' such that if $h = \text{CHash}(\text{ck}, \text{vk}, m, r) = \text{CHash}(\text{ck}, \text{vk}, m', r')$.

A.2 Threshold encryption and signature security proofs

Theorem 5. *The signature scheme Π_{SUF} has selective unforgeability under the MSIS and MLWE assumptions.*

Proof. We prove this theorem by a reduction to a MSIS instance. Let $\mathcal{A}$ be an adversary with advantage ε to forge a signature for Π .

Game $\mathbf{G}_0$: the original game

- The adversary chooses a message m^* .
- The challenger runs $\Pi.\text{KeyGen}$ to compute $\text{vk} = (A, B = AR, B', u)$, $\text{sk} = R$ where $R \leftarrow \$ S_1^{(n+m) \times \ell n}$ and $A \leftarrow \$ R_p^{n \times (n+m)}$, $B' \leftarrow \$ R_p^{n \times \ell n}$, $u \leftarrow \$ R_p^n$. It sends vk to the adversary.
- The adversary has access to a signature oracle $\mathcal{O}_{\text{Sign}}$ for messages $m \neq m^*$.
- The adversary outputs a signature σ^* .

Reduction setup. The reduction $\mathcal{R}$ takes as input a MSIS instance A , and the message m^* that $\mathcal{A}$ wants to forge a signature on. It computes $B' \leftarrow AR'$ where R' is sampled in the same way as R . Under the MLWE assumption, the distribution of AR' is indistinguishable from uniform, and the adversary cannot distinguish. It computes $B \leftarrow AR - m^*G$. Again under the MLWE assumption, AR is indistinguishable from uniform, thus $AR - m^*G$ is statistically indistinguishable from AR . It samples $s^* = (s_1^*, s_2^*, s_3^*)$ a gaussian vector and sets

$$u = [A \ AR \ AR'] \begin{bmatrix} s_1^* \\ s_2^* \\ s_3^* \end{bmatrix} \mod p.$$

It sends the public key (A, B, B', u) to $\mathcal{A}$.

To answer oracle queries on any message $m \neq m^*$, the reduction computes signatures using the trapdoor R with gadget matrix $(m - m^*)G$ [22]. To answer

oracle queries on m^* , R is not a valid trapdoor hence the reduction returns (s_1^*, s_2^*, s_3^*) . This game is statistically indistinguishable from the original game as this follows the same gaussian distribution.

Main reduction. Assume that $\mathcal{A}$ outputs a forgery $\sigma = (s_1, s_2, s_3)$ for m^* . Then, we have

$$\begin{bmatrix} A & AR & AR' \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} A & AR & AR' \end{bmatrix} \begin{bmatrix} s_1^* \\ s_2^* \\ s_3^* \end{bmatrix} = u \pmod{p}.$$

Then, $v = s_1^* - s_1 + R(s_2^* - s_2) + R'(s_3^* - s_3)$ is a solution to the MSIS instance A if $v \neq 0$ i.e., $\sigma \neq s^*$. This is the case with overwhelming probability since s^* was chosen independently by the reduction at the beginning of the game.

Theorem 6. *The threshold decryption scheme from Section 5 has INDCCA2-TD security under the IND CPA security of the Regev encryption and the soundness and zero-knowledge properties of the ZKP. More precisely, for any adversary $\mathcal{A}$ that makes at most q decryption queries, we have*

$$\text{Adv}_{\text{INDCCA2-TD}}(\mathcal{A}) \leq \frac{\text{Adv}_{\text{INDCPA}}(\mathcal{A}) + q \cdot \text{Adv}_{\text{SND}}(\mathcal{A})}{1 - \text{Adv}_{\text{ZK}}(\mathcal{A})}$$

Proof. We prove this theorem by a reduction to the IND CPA security of the classical Regev encryption. Let $\mathcal{A}$ be an adversary with advantage ε to break the INDCCA2-TD security of the threshold decryption scheme.

Reduction setup. The reduction $\mathcal{R}$ takes as input a Regev public key pk . For a corruption request on identity i , the reduction samples sk_i at random and sends it to the adversary. Under the generalized Shamir's secret sharing perfect secrecy, the adversary has no advantage to distinguish this from an honest decryption key. To answer a decryption query on a ciphertext (c, π) , the reduction uses the extractor from the ZKP to recover the plaintext m encrypted in c . Let q be the number of decryption queries, this fails with probability $q \cdot \text{Adv}_{\text{SND}}(\mathcal{A})$.

Main reduction. When $\mathcal{A}$ sends its target m_0, m_1 , $\mathcal{R}$ sends the same to its challenger. Upon receiving the challenge c^* , the reduction sends (c^*, π^*) to the adversary by using the simulator of the ZKP to generate π^* . The adversary has advantage $\text{Adv}_{\text{ZK}}(\mathcal{A})$ to distinguish this from an honest ciphertext. Finally, if the adversary manages to make a correct guess, so does the reduction, hence the probability for the reduction to answer correctly is

$$p_{\mathcal{R}} \geq \varepsilon \cdot (1 - \text{Adv}_{\text{ZK}}(\mathcal{A})) - q \cdot \text{Adv}_{\text{SND}}(\mathcal{A}).$$

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